Monash University
FIT5147 Data Exploration and Visualisation
Semester 1, 2026

Data Exploration Project (DEP)

DEP Part 1: Project Proposal
DEP Part 2: Project Report

You are asked to explore and analyse data using visualisation about a topic of your choice. It is an **individual assignment** and **worth 35%** of your total mark for FIT5147. The Data Exploration Project (DEP) has two parts:

- **DEP Part 1** - Weight: 2%. Submission: Project Proposal.
- **DEP Part 2** - Weight: 33%. Submission: Project Report.

Relevant Learning Outcome

- Perform exploratory data analysis using a range of visualisation tools (limited to a choice of either Tableau or R)

Generative AI tools can be used to help with the written and code content of this assessment task, but this must be clearly declared in your report. See the Originality section of this document for further details.

Overview of the Assessment Tasks

The following table describes the overall tasks and the scope of Part 1 and Part 2.

Scope	Task
DEP Part 1 (2%)	1. Identify the project topic, some related questions that you want to address, and the data source(s) that you will be using to answer those questions. Outline these in your **Project Proposal**. 2. Submit Project Proposal in the Moodle Assessments section at the end of **Week 2**. 3. Discuss the proposal with your tutor **in your Week 3 Applied Session** (*after* the submission in Moodle) and wait for approval from your tutor before proceeding further. Do not seek approval from the lecturer. *The proposal is a hurdle requirement. You must get your proposal approved in order to continue with Part 2. See the Late Submission section for details or if you joined the unit late.*
DEP Part 2 (33%)	4. Collect data and wrangle it into a suitable form for analysis using whatever tools you like (e.g., Excel, R, Python). 5. Explore the data visually to answer your questions and/or to find other interesting insights only using **Tableau** or **R**. The exploration must rely on visualisations and visual analysis from those technologies. You can use analytical methods or statistical analysis where appropriate. 6. Write a **Project Report** detailing your findings and the methods that you used. This must include properly captioned figures demonstrating your visual analysis (i.e., your visualisations must be referred to correctly in your report). 7. Submit your Project Report in **Week 7**.

Read the rest of this document before deciding on your project topic, as the proposal is for the entire **Data Exploration Project** and the **Data Visualisation Project**, which is the second major assignment of this unit. See the end of this document for an example proposal and potential data sources to get yourself started. Be careful not to copy this proposal; it is an example proposal, not template text.

Choosing a Topic and Data

The choice of topic, data, and the questions you seek to answer should allow for interesting and detailed analysis in the Data Exploration Project (DEP) and the subsequent Data Visualisation Project (DVP, due at the end of semester), which involves presenting the findings from your DEP in a specifically designed narrative interactive visualisation format.

Ask appropriate questions

Good questions are general and not linked to specific parts of the data, allowing for more open-ended and exploratory analysis. This will then allow you to demonstrate your skill at discovering insights from data using appropriate data visualisation practice. For instance, asking "*Where are the safer parts of the network?*" is a <u>good question</u> that lets you explore various interpretations of how to link terms like "where" and "safety" to the data about a network, doesn't expect a single answer and doesn't limit the data to specific values. In contrast, "*Which suburb had the lowest value of number-of-deaths in 2024?*" is <u>**not** a good question</u> as it is specific to the data (names a specific field and year and how to interpret "where"), is easy to answer with one visualisation (only wants to know the lowest value, not trends or ranks or comparisons) and therefore limits the exploration and visualisation possibilities.

It is strongly recommended that you avoid questions that are:

- too easy to answer (e.g., "What is the correlation between x and y?", "What is the average value of z variable?", "What are the top/bottom N values?"), or
- too difficult to answer (your work would take longer than the time available in the unit), or
- not relevant to the unit (e.g., training a machine learning model or making predictions), or
- not possible to answer from the available data (e.g., determining if A causes B is often not possible from correlation analysis alone).

Proposals with such questions will be rejected. If you are in doubt, talk to the teaching staff during face-to-face teaching times or ask for confirmation on Ed.

Collect appropriate data

How do you know if you have appropriate data? This depends on your topic and questions. You should ensure your data is big enough, i.e., has enough breadth and depth to invite interesting exploration. Combining data from different data sources is an ideal way to help add to the originality of the topic. To encourage different visualisation techniques, your data will likely have a mixture of different data types. Time series (whether this be aggregated or detailed, such as months and years, or milliseconds) may be useful for your topic, and spatial, relational or text-based data add useful complexity. If in doubt, talk to teaching staff during face-to-face teaching times or in a consultation before the due date.

The chosen topic should be topical and some of the data must be recently collected, ideally from the last two or three years. The **data must be accessible to the teaching staff**, so the use of open data is encouraged (see the list of suggested data sources at the end of this document). Use of closed or proprietary data is allowed as long as explicit permission for use in this assignment is granted by the original authors or copyright holders. If you have closed data, you must still make it available to your teaching staff to access, i.e., via a shared Google Drive. Simulated data is not allowed.

Avoid common topics. Common topics including COVID-19, Netflix, AirBnB, car accidents, crime, house sales, car sales, world cup soccer, or electric vehicle sales should be avoided. Topics similar to the proposal example at the end of this document, i.e., traffic accidents and poor weather, must also be avoided. If you do have personal motivation for any of these mentioned common topics, you will need to propose a completely new angle to exploring the theme through novel questions with a mixture of new data sources. It is highly recommended to discuss your intentions with the tutor of your Applied Session prior to the proposal submission to avoid immediate rejection of the proposal.

DEP Part 1: Project Proposal (2%)

Write a Project Proposal, in a **one-page** PDF document consisting of the following sections:

1. **Project Title**
   A descriptive title for your project.
2. **Topic Introduction**
   One paragraph introducing the topic. This should include why it is a topical subject (for example, has it been in the news recently), and who might benefit from the insights you seek from your questions.
3. **Motivation**
   One paragraph describing why you personally are motivated to study this topic.
4. **Questions**
   Three questions you wish to answer using the data.
5. **Data source(s)**
   Briefly describe the data source(s) you will use. This should include direct URLs of data source(s) and a description for each source: what is the data about, what is the size of the data (e.g., number of rows, number of columns), the type of data (e.g., tabular, spatial, relational, or textual), the type of attributes (e.g., categorical, ordinal, etc.) and the temporal intervals and period (e.g., monthly records between 2019 and 2023). Explain how the data will contribute to the exploration.
6. **References**
   The bibliographical details of any references you have cited in the previous sections.

Include your full name, student ID, tutor names, and Applied Session class number in the document header or footer. There should be **no cover page.**

DEP Part 2: Project Report (33%)

Write a Project Report with the following structure:

1. **Introduction** (no more than 1 page)
   Project title, topic detail, problem description, brief motivation, and questions.
2. **Data Wrangling and Checking** (no more than 3 pages)
   Description of the data and data sources with URLs of the data, the steps in data wrangling (including data cleaning and data transformations) and tools that you used. Description of the data checking that you performed, errors that you found, your method and justification for how you corrected errors, and the tools that you used. A comprehensive checking process is expected to justify data correctness, even if the dataset is believed to be clean.
3. **Data Exploration** (at least 5 pages)
   Description and justification of the data exploration process followed and the choice of visualisations (including figures and descriptions of findings) and statistical tests (if applicable) you used, what you discovered, and what tools you used and why.
4. **Conclusion** (around half a page)

Summary of what you learned from the data and how your data exploration process answered (or didn't answer) your original questions.

5. **Reflection** (around half a page)
   Brief description of what lessons you learnt about data visualisation in this project and what you might have done differently in hindsight.
6. **Bibliography** (not included in the page limit)
   Appropriate references and bibliography (this includes acknowledgements to online references or sources that have influenced your exploration and that you have cited) using either the APA or IEEE referencing system.

Include your full name, student ID, tutor names, and Applied Session class number on a cover page, or in the header or footer of the first page.

The report should be **not longer than 10 pages for all sections mentioned above**, excluding cover page, table of contents, bibliography and appendix. Your report will be the sole basis for judging the quality of the data checking, data wrangling, data exploration, as well as the degree of difficulty. Thus, include sufficient information in the report. It should, for instance, contain images of visualisations used for exploration and the results of any statistical analysis. You should include any analysis that you carry out even if it is incomplete or inconclusive as it demonstrates that you have thoroughly explored the dataset. No programming code needs to be submitted but you must be able to supply any code or workbooks you used in any part of your assignment work when requested by teaching staff.

If you wish to provide additional material, an **Appendix** of up to 5 pages may be added at the end of the document. However, the Appendix will not be marked. Therefore, you should only use it to provide supplementary material that is not essential to the report or the reader's understanding. Be sure to clearly title this section as Appendix.

Marking Rubric

DEP Part 1: Project Proposal (2%)

- ***Completeness and Timeliness*** [1%]: All components of the Proposal are included and it is submitted on time.
- ***Suitability and Clarity*** [1%]: Motivation, Questions and Data Sources.
    - **Motivation**: A well-formulated project description with detailed information; a compelling and worthwhile topic to explore and visualise as a real-world problem.
    - **Questions**: Three well-crafted questions that can be clearly answered through data visualisations. Each question requires sophisticated analysis of relationships and patterns across multiple attributes and demonstrates potential for innovative visualisation approaches to reveal insights and complex patterns.
    - **Data Sources**: A clear description of data sources and datasets, including justification for which questions you will answer with each. The data must be sufficiently large or complex to require exploration and analysis. All datasets must be easily available, with URLs provided directly to all data files used, not just a link to a webpage for the dataset, so your marker can easily download the data from the source. For private and proprietary data, links to the data and evidence of permission to use it must still be provided. If data is to be scraped, you will need to provide details about where it will be scraped from, what software or service will be used to scrape it and samples of the resulting expected data.

The marking of your proposal is only based on the work you submit at the time of the deadline. However, you cannot work on Part 2 of your DEP until your Part 1 proposal has been fully approved. After your proposal is submitted, you will meet with your tutor during the Week 3 Applied Session to discuss your Project Proposal, receive verbal feedback and ideally approval to start. Written feedback will not be provided so attendance is vital. If your proposal is rejected, your tutor will specify the reasons and suggest areas for improvement. You will need to make these amendments to your proposal, resubmit it on Moodle, and get it approved by your tutor prior to commencing your DEP Part 2 work. For this reason, you are encouraged to submit a proposal on time so your tutor can provide feedback ASAP.

DEP Part 2: Project Report (33%)

Criteria	Below 50%	Pass (50%+)	Credit (60%+)	HD (80%+)
Data Complexity, Wrangling, Checking and Cleaning (7%)	Inappropriate checking, cleaning, or wrangling. 0 if no demonstration of data checking and cleaning.	Appropriate data cleaning and checking. Demonstrated ability to get data into R or Tableau.	Good choices and clear justifications for error checking, cleaning and transforming of non-tabular data (e.g. spatial, relational, textual); large datasets (observations or dimensions) and/or multiple datasets.	Excellence in data processing demonstrated and documented. Evidence of significant complexity in the wrangling, cleaning, transformation, or data collection (e.g. scraping).
Data Visualisation and Design Choices (9%)	No visualisations; unsuitable or poor choice of visualisations; pixelated / poor quality images or illegible visualisations. 0 if not using Tableau or R.	Suitable visualisations, which are well presented, described, readable and interpretable.	Visualisations are appropriate for the intended purpose; appropriate labeling of axes and visualisations; clear legends when needed; saliency of patterns and trends.	Variety of high-quality, complex and/or creative visualisations with high attention to detail. Clearly justified design choices including visualisation idioms, choice of visual variables, layout and labelling.
Analytical Methods and Interpretations of Data and Topic Questions (9%)	Unsuitable analysis or misinterpretation of the data and the topic's questions. 0 if no data analysis is demonstrated.	Demonstrated suitable analysis and interpretation of the data and topic questions.	Analysis that is appropriate for the intended purpose; justification and explanation of the exploration process and use of statistical measures; identification of trends, patterns, and insights.	High quality of visual analysis demonstrated. Sophisticated and correctly used analytical methods such as clustering; dimensionality reduction; sophisticated aggregation and/or filtering; non-linear model fitting; correct use of statistical tests; or complex time series analysis.
Written Report: Quality and Completeness (8%)	Poor report, or missing sections.	Good report with logical structure with all the expected sections: Introduction, Data Wrangling, Data Checking, Data Exploration, Conclusion, Reflection, Bibliography. Referencing of sources, figures and tables. Correct grammar and spelling.	High quality of writing and figures/images with minimal errors. Correct referencing of figures and tables within the text, and correctly used academic referencing of sources.	Professional report with excellence of writing combined with high quality figures/images. Clearly articulated findings; awareness of limitations; deep exploration; thorough conclusions.

Originality

Since this is academic work, it must be original and clearly distinguish between your own contributions and those based on other's work. If you include data, facts, opinions or any other written or graphical information from another source, you must cite and reference it according to the APA or IEEE style guide. Otherwise this may be considered **plagiarism**. Direct quotations or reproductions must also adhere to the appropriate APA or IEEE style.

For the same reason, you must **declare any software or third party service** you use to write or code your submitted work, including the use of **Generative AI tools**, such as ChatGPT, Grammarly and Google Translate. You must declare the tool(s) and prompt(s) you used at the end of your report (this is not included in the page limit). It is highly recommended that you thoroughly check and validate any generated material that you include or use as a basis of any of your assignment. Remember that AI-generated material can be biased, simplified or false and is often not high enough quality for this assessment.

In your report you are encouraged to repeat the questions from your proposal. **This is the only self-plagiarism that is allowed**. To ensure originality of your work:

- your topic and dataset cannot have been used in any other unit.
- you may not reuse any code or written content or data from previous assessment tasks for any unit.
- content from previous assignments or sample reports cannot be used.
- if **retaking this unit from a prior semester**, you must choose a **completely new topic and dataset**.

If concerns arise regarding the originality of your work – whether due to plagiarism, collusion, contract cheating, the use of unapproved software or the use of Generative AI tools without declaration – your academic integrity will be reviewed. Confirmed breaches of academic integrity may result in penalties affecting your assignment mark, this unit, or even your enrolment. If you need further information on academic integrity at Monash, we recommend you review the academic integrity module.

Submission and Due Dates

Once you have completed your work, take the following steps to submit your work.

1. Save your proposal or report as a PDF document.
2. Name your file using the following structure**: Proposal_Surname_StudentID.pdf** or **DEP_Surname_StudentID.pdf**
3. Submit and upload your document.

- **Project Proposal**: Submit the **PDF** file at the end of **Week 2**, for feedback during **Week 3**.
- **Project Report**: Submit the **PDF** (see requirements above) in **Week 7**.

The exact dates and times of the deadline are on Moodle.

Your assignment must show a status of "Submitted for grading" before it can be marked. Any submission in "Draft" mode will not be marked.

Late Submissions

- There will be **zero marks for late Project Proposal submissions**. Everyone must submit the Project Proposal. Even if the deadline has passed, you must still submit a proposal (with a grade of 0) as your project **must be approved** before you can continue working on the Data Exploration Project. The proposal is a hurdle requirement. If it is not submitted and approved by your tutor, the mark for the entire Data Exploration Project (Parts 1 and 2) is 0.

- For the Project Report, submissions received after the deadline (or after an extended deadline for those with an extension or special consideration) will be **penalised at 5% of the total available mark [33%] per calendar day up to a maximum of 7 days.** If submitted after 7 days, it will not be marked, will receive zero marks and no feedback will be provided. This will also impact your DVP work as the two major projects relate to each.
- For further information on eligibility for **Extensions or Special Consideration,** see: https://www.monash.edu/students/admin/assessments/extensions-special-consideration

Example Data Sources

The following is a list of data sources to get started. Feel free to use these as a source of inspiration and ideas for your project. You are not limited to the data sources listed below.

- Data search tools and repositories, e.g.:
  - Google dataset search: https://toolbox.google.com/datasetsearch
  - Google Trends: https://www.google.com/trends/explore
  - Google Ngram Viewer: https://books.google.com/ngrams
  - Registry of Open Data on AWS: https://registry.opendata.aws/
  - Kaggle: https://www.kaggle.com Note that using data from Kaggle exclusively is not acceptable, you must use at least one additional data source.
  - Science Hack Day: http://sciencehackday.pbworks.com/w/page/24500475/Datasets
- Open local and national government data portals, e.g.:
  - Victorian Government Data: http://data.vic.gov.au/
  - Australian Government Data: http://data.gov.au/
  - Digital Atlas of Australia: https://digital.atlas.gov.au/search
  - Australian Bureau of Statistics: https://www.abs.gov.au/statistics
  - Atlas of Living Australia https://ala.org.au/
  - European Union Open Data: https://data.europa.eu/en
  - UK Government Open Data: https://data.gov.uk/
  - U.S. Government Open Data: https://www.data.gov/
- Humanitarian and environmental data sources, e.g.:
  - UNdata: http://data.un.org/
  - The World Bank Data Catalog: https://datacatalog.worldbank.org/
  - Our World in Data: https://ourworldindata.org/
  - Berkeley Library Health Statistics: http://guides.lib.berkeley.edu/publichealth/healthstatistics/rawdata
  - Movebank (animal tracking, some open data), https://www.movebank.org/
- Open corporate/industry data, e.g.:
  - Inside Airbnb: http://insideairbnb.com/get-the-data.html (Note that Airbnb is not an approved topic alone, see Avoid Common Topics)

Example Project Proposal

Please note this mock example is relatively old now. We expect your data to include some more recent data, i.e., data from 2023, 2024, 2025 or even 2026. It is possible to complete this example project with only Data Source A and B, but data from C provides different opportunities and additional difficulty when doing the exploration and visualisations. If done well, this added depth and difficulty can gain extra marks but might take longer to complete. The student could use both datasets A and B to identify temporal aspects in the data, such as accidents near to sunset and sunrise across the whole dataset, but dataset C allows them to identify areas which are poorly lit and see if this correlates with the spatial pattern of pre-sunrise and

post-sunset accidents. Furthermore, whilst Data Sources A and C are currently tabular data, they can be converted to spatial features and spatial analysis can be carried out.

Name: Jesse van Dijk, **Student ID:** 12345678, **Teaching Associate:** Jo Bloggs & Alex Smith, Applied 01.

Project Title: Causes of Serious Bicycle Accidents in Canberra

Introduction

Recent media and industry reports indicate that Australian roads are becoming even more dangerous for cyclists [1,2]. I believe this is an important topic for many audiences such as cyclists, road safety officers, and public health policy makers. Therefore I want to find out more about the factors that affect bicycle accidents in Canberra.

Motivation

I am a keen cyclist and am concerned about cycling in Australia. I have recently moved to Canberra from the Netherlands where cycling is very safe and accidents linked with road vehicles is unusual. I have noticed it is difficult to see during sunset on a number of roads and would like to see if this pattern is evident in the data.

Questions

1. What are the more common kinds of serious bicycle accidents in Canberra, and how do these vary over different time periods (e.g., hour of day/day of week/month/season)?
2. Is there a relationship between the lighting conditions and these accidents?

Data sources

A. **ACT Road Cyclist Crashes** 2012 to 2021**,** which have been reported by the Police or the Public through the AFP Crash Report Form. This data is tabular data: ~1K rows × 11 columns. It has both spatial and temporal attributes including the geographical (latitude and longitude) location and a datetime stamp for the time of accident. Some numerical and simple text attributes relating to the incident. i.e. number of casualties, description of accident, including direction of traffic. (https://www.data.act.gov.au/Justice-Safety-andEmergency/Cyclist-Crashes/n2kg-qkwj)

B. **Canberra's sunrise and sunset times**, 2012 to 2021. Tabular data in HTML: ~365 rows × 4 columns for each year to be scrapped from sunrise website. Columns are simply date, time of sunrise, time of sunset and hours of daylight. https://sunrise.maplogs.com/canberra_act_australia.331491.html?year=2021

C. **ACT Streetlights, 2021**. Tabular data in CSV format with ~80K rows × 10 columns. These include latitude and longitude for the streetlight location and various text columns including lamp type, Luminaire, height and street and suburb name. There is no date column for the age of the lamp, but the source of the data is dated from 2017 and was last updated in Nov 2021. https://www.data.act.gov.au/Infrastructure-and-Utilities/ACT-Streetlights/cfpr-4tpw/about_data

Data Source A will be used to address Question 1, whilst A to C will allow me to answer Question 2.

References

[1] Guthrie, Susannah (2020), *Report shows 'alarming spike' in cyclist deaths on Australian roads*, Car Advice, 05.08.2020. URL: https://www.caradvice.com.au/870483/cyclist-deaths-australia/ [Accessed on: 23.07.2021 (checked 25.02.2024)].

[2] Australian Automobile Association (2020), Benchmarking the Performance of the National Road Safety Strategy, July 2020, URL: https://www.aaa.asn.au/wp-content/uploads/2020/07/AAA_QBR_June_2020_Final_web.pdf [Accessed on: 23.07.2021 (checked 25.02.2024)].